



Coachella 2026 Live Music Festival

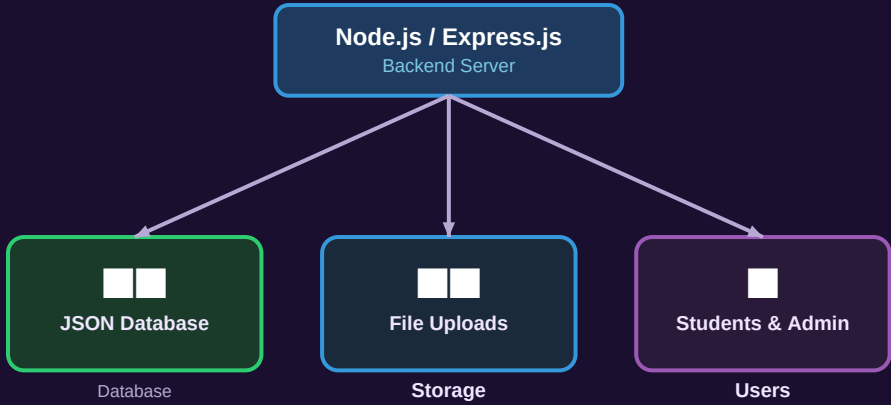
Ultimate Digital Festival Experience | April 10–12, 2026 • Indio, California

PROJECT BASED LEARNING FROM FWD

Abstract

The **Coachella 2026 Live Music Festival** is a full-stack interactive web platform where festival-goers can browse the lineup, book tickets, explore food & merchandise, and manage their festival experience end-to-end. Developed using **HTML, CSS, JavaScript, Node.js, and Express.js**, the platform features a real-time countdown timer, smart artist search powered by a **Trie data structure**, a trending algorithm using a **Min-Heap**, custom QR-code ticket generation, an interactive festival map, a full e-commerce cart with checkout flow, user authentication, and a live admin dashboard with analytics. The system employs a **HashMap** for O(1) artist lookups and a **Graph** to model stage adjacency. All data is managed client-side with JSON, making the app lightweight and deployment-ready.

System Architecture



— DSA CO-WISE IMPLEMENTATION —

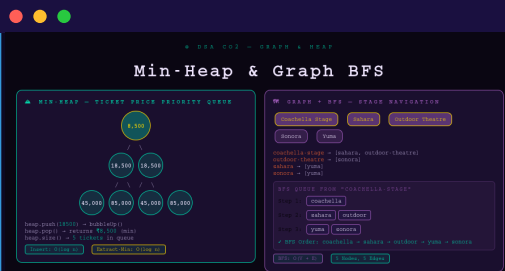
CO1 Linear & Non-Linear Data Structures

- ✓ HashMap: O(1) artist lookup via hash function
- ✓ Hash(key) = $\sum(\text{charCode} \times 31^i) \% \text{tableSize}$
- ✓ Trie: Autocomplete for artist name search
- ✓ Insert/Search in O(L) where L = word length
- ✓ 40 artists indexed; prefix search in real-time



CO2 Trees, Heaps & Graph Algorithms

- ✓ Min-Heap: Ticket price priority queue
- ✓ BubbleUp/SinkDown for heap property
- ✓ Graph: Stage adjacency (5 nodes, 5 edges)
- ✓ BFS traversal for stage-to-stage navigation
- ✓ Insert O(log n) | BFS O(V+E) complexity



CO3 Sorting & Searching Algorithms

- ✓ Heap Sort: Artists ranked by favorite count
- ✓ Heapify → extract-max → sorted leaderboard
- ✓ Binary Search: Ticket price lookup in cart
- ✓ lo/hi/mid pointer logic on sorted array
- ✓ HeapSort O(n log n) | BinarySearch O(log n)



CO4 Applied DSA & Algorithm Design

- ✓ QR Code: LCG hash → 21×21 canvas grid
- ✓ 3 finder patterns + data modules drawn
- ✓ Running Sum: Prefix array for revenue chart
- ✓ prefix[i] = prefix[i-1] + daily[i]
- ✓ QR O(n²) canvas | Running Sum O(n) space



Conclusion & Future Scope

Modern Web Technologies

Secure & Scalable

DSA-Powered

Mobile-First

Deployment Ready

Modern Web Technologies • Secure & Scalable

HTML5 • CSS3 • JavaScript ES6+ • Node.js • Express.js • JSON DB

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